TARUN MUNDHRA | 23EX10038

B.S.(Hons.) in EXPLORATION GEOPHYSICS

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	4YRS B.S	IIT Kharagpur	8.20 / 10
2023	CLASS XII, CBSE	Guru Nanak Public School, Sambalpur	91.6%
2021	CLASS X, CBSE	Aditya Residential School, Sambalpur	96.8%

INTERNSHIPS

Quick Dry Cleaning Software | Software Development Intern **[May 2026 - Nov 2026]**

- Built an AI-first, **multi-tenant** B2B SaaS backend with **PostgreSQL RLS** to enforce tenant isolation & **prevent cross-tenant data leaks**
- Designed custom **NestJS IAM guards** enforcing territory-based API access, closing gaps standard RBAC libraries couldn't handle
- Architected an **LRU cache** for sub-ms API permission checks alongside **Kubernetes** pre/post-**DB migration** Jobs for zero downtime
- Engineered real-time LiveKit **voice AI agents**, utilizing **Temporal.io** to guarantee **fault-tolerant** execution of multi-step LLM tasks
- Designed **event-driven background worker queues** with BullMQ/Redis to handle async billing, webhook dispatch & notifications

AMRR TechSols Pvt Ltd | Software Development Intern **[May 2026 - Jul 2026]**

- Engineered the foundational core **AI risk engine** for the group's legal-tech platform, deployed at enterprise scale in production
- Developed production **RAG** pipeline using **LangChain** with **BM25 reranking**, **LLMOps** scoring layer, and hallucination-aware **evals**
- Devised a hybrid semantic search engine using **pgvector's HNSW indexing** to achieve **sub-50ms** legal document retrieval at scale
- Improved native **document conversion quality by 70%** with precise visual layout alignment using PyMuPDF and parallel **LibreOffice**
- Engineered end-to-end **PII redaction pipeline** with AES-256-GCM encryption, privacy-by-design, and GDPR-ready architecture

PROJECTS

KGP Presence- AI-Powered Campus Attendance App | React Native, MongoDB | Lead Developer **[Jan 2026 - Aug 2026]**

- Shipped an **Expo app** to **Google Play Store** (100+ installs) via **EAS** Build; deployed its **AWS Lambda** backend via **GitHub Actions CI/CD**
- Built **OCR timetable extraction** via **Gemini API** and event creation through voice command using **Sarvam STT** with **Groq LLM** parsing
- Built real-time attendance analytics dashboard with **TanStack Query zero-latency caching**, smart event alerts, and **offline data sync**
- Architected serverless **Node.js/MongoDB** backend with secure **JWT** auth, **robust bcrypt** hashing, and OTP-based email verification

High-Frequency Matching Engine & Market Simulator | Go, Concurrency, Redis, TimescaleDB **[Mar 2026]**

- Built a **low-latency LOB matching engine** in Go with price-time priority, using **sync.Pool and per-symbol sharding** to scale throughput
- Engineered **advanced order types** (Iceberg, Pegged, **Stop-Limit**) and Self-Trade Prevention (**STP**) to eliminate abusive wash trading
- Built **automated market make**r and noise trader agents to simulate book liquidity and retail order flow with **stochastic GBM model**
- Architected a real-time data pipeline streaming Level-2 LOB depth snapshots and trade feeds via **Redis Pub/Sub** and **WebSockets**

Low-Latency AI Video Streaming Engine | Python, FastAPI, FFmpeg, HLS, Cloud Object Storage, Asyncio **[Aug 2026]**

- Architected a **progressive AI video pipeline**, decoupling scene generation from **live HLS chunk slicing** to cut **TTFF latency by 7x (<3s)**
- Engineered an async manifest holding pattern, eliminating **client polling race conditions** and severe FastAPI event-loop blocking
- Implemented dynamic **PTS** timestamp synchronization across scene boundaries, preventing browser **MSE decoder buffer stalls**
- Designed an immutable storage and edge caching layer with tenant-scoped access policies, **eliminating redundant cloud reads**

ENTREPRENEURIAL EXPERIENCES

JuryAssist (Product by Veritos) | Founding Engineer **[Aug 2026 - Present]**

- Spearheaded 0-to-1 architecture of an **agentic AI** UK legal platform, scaling adoption to **10,294 active users** across top law firms
- Drove product-led growth strategy platform-wide, boosting **user adoption** and premium-tier conversions across the client base
- Built **zero-knowledge AI platform** with 6-LLM consensus, distributed **state locking &HMAC**-verified **idempotent payment webhooks**

SKILLS AND EXPERTISE

Languages &Frameworks:TypeScript | Python |C++ | Go |SQL | Nest.JS |FastAPI | Express.js |React.js |Next.js | Expo | LangChain
Databases, Storage &Cloud: PostgreSQL (pgvector) | MongoDB | Redis | TimescaleDB | AWS | Supabase | Temporal.io | BullMQ
Tools &Security: Docker | Kubernetes | Github Actions | JWT | RBAC/IAM | SQLAlchemy | Kafka | HMAC | Vitest | Linux/Bash

POSITIONS OF RESPONSIBILITY

Vice-Captain Interhall GC OpenSoft Competition | MMM Hall of Residence **[Mar 2026]**

Led a **20-member team** to architect and deliver a **high-throughput** trading simulator under competitive inter-hall conditions

COURSEWORK INFORMATION

Programming &Data Structures, Linear Algebra, Probability &Statistics, Advanced Calculus, **Pattern Recognition**, Partial Differential Equations, Economics, Project Engineering and Management, Engineering Design Process**(EDP)**, Geophysical **Inverse Theory**

EXTRA CURRICULAR ACTIVITIES

- **ET AI Hackathon 2.0** (**The Economic Times**): Shortlisted for **Phase 2**: Build Sprint among thousands of leading national participants
- Completed **NCC B Certificate** and **certified training camp** involving **intensive drill** alongside **leadership and teamwork skills development**

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